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23 Oct 1973
SUPERSEDING
MIL-O-38205 (USAF)
6 April 1964

MILITARY SPECIFICATION

OVENS, FOOD WARMING, ELECTRIC
HDU-2/B AND HDU-23/B

1. SCOPE

1.1 Scope. This specification covers two types of electrically heated, forced air circulating food warming ovens, utilized for reconstituting frozen precooked meals, partially cooked food and/or the heating of canned foods.

1.2 Types.

Type I - 115/200 volt AC, 400 Hz, 3 phase, 4 wire, "Y" connected, grounded neutral.

Type II - 28 volt DC, 2 wire, grounded electrical power system.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/2	Aluminum Alloy 3003, Plate and Sheet
QQ-A-250/8	Aluminum Alloy 5052, Plate and Sheet
QQ-B-654	Brazing Alloy, Silver
QQ-S-763	Wire, Steel, Corrosion Resisting
QQ-S-766	Steel Plates, Sheets, and Strip, Corrosion Resisting
ZZ-R-765	Rubber, Silicone, Low and High Temperature and Tear Resistant
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-W-5088	Wiring, Aircraft, Installation of

FSC 7310

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MIL-W-8604	Welding of Aluminum Alloys: Process for
MIL-S-12204	Solder, Aluminum Alloy
MIL-M-13966	Meal, Precooked, Frozen

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-882	Systems Safety Program, for Systems and Associated Subsystems and Equipment; Requirements for

(Copies of Specifications, Standards, Drawings and Publications required in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

Occupational Safety and Health Act (OSHA) Standards with Supplemental Revisions

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Preproduction. This specification provides for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Safety. All equipment furnished under this specification shall be analyzed for category III and category IV hazards as defined in MIL-STD-882 and shall be consistent with "OSHA" standards.

3.4 Materials. Materials that are not covered by applicable specifications or that are not specifically described herein shall be of the best quality, shall be of the lightest practical weight, and shall be suitable for the intended purpose.

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3.4.1 Fungus-proof materials. All materials shall be vermin, fungus, and insect-proof or shall be treated to be vermin, fungus, and insect-proof. Fungus-proof materials shall not impart objectionable flavors or odors to food and shall be nonhygroscopic.

3.4.2 Metals. All metals shall be of the corrosion-resistant type or shall be suitably treated to resist corrosion due to fuels, salt spray, or any other atmospheric conditions likely to be met during storage or normal service.

3.4.3 Hardware. All hardware such as door hinges, latches, and handles shall be corrosion-resistant, smooth finished, fabricated from solid material with integral or plated finish, designed for ease of cleaning, and secured to the oven in such a way that it can be replaced easily when broken or worn.

3.4.4 Outer case. The outer case shall be constructed from aluminum alloy sheet of at least 18 gage and conforming to QQ-A-250/2 or QQ-A-250/8.

3.4.5 Covering and mounting panels. Panels used to cover the insulation and to serve as mounting panels for the component parts shall be fabricated from aluminum alloy sheet conforming to QQ-A-250/2 or QQ-A-250/8.

3.4.6 Heating chamber. The heating chamber of the oven shall be constructed of stainless steel sheet conforming to class 302 or 304, condition A (annealed) of QQ-S-766.

3.4.7 Heater baffle. The heater baffle shall be constructed of stainless steel sheet conforming to class 302 or 304, condition A (annealed) of QQ-S-766.

3.4.8 Baffle jacks. The baffle jacks shall be constructed of nickel silver.

3.4.9 Drop-leaf door. The outer cover of the drop-leaf door shall be constructed of 18 gage aluminum alloy sheet conforming to QQ-A-250/2 or QQ-A-250/8. The inner liner shall be constructed from stainless steel conforming to class 302 or 304, condition A (annealed) of QQ-S-766.

3.4.10 Door gasket. The door gasket shall be fabricated from silicone rubber extrusion conforming to ZZ-R-765 and shall be capable of withstanding and remaining functional at a temperature of plus 475° F. The silicone rubber door gasket shall be constructed of four sections to facilitate disassembly and cleaning. A flat stainless spring shall be placed in the silicone gasket, and the end of the spring steel is to hook into the front top and the front bottom of the outer case. The door shall close tightly against the silicone case.

3.4.11 Insulation. Materials used to insulate the heating chamber from the rest of the oven shall be of both rigid and nonrigid types and shall be sanitary and odorless (see 3.4.1), inorganic, nonhygroscopic, noncorrosive, and shock-resistant. The insulating materials shall have a low "K" factor, long life, and be of a non-packing type. The insulating materials shall be of sufficient quantity and quality to enable the oven to meet the performance requirements specified herein.

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3.4.12 Food rack. The food rack shall be constructed of 0.125 diameter stainless steel wire conforming to composition 302 or 304 of QQ-S-763.

3.4.13 Drip pan and auxiliary food tray. The drip pan and auxiliary food tray shall be constructed of 18 gage aluminum alloy sheet conforming to QQ-A-250/2 or QQ-A-250/8.

3.4.14 Solder. All solder used in the interior of the heating chamber shall withstand a temperature of 475° F. without melting. The solder shall not be toxic and shall conform to type I or II, group A of MIL-S-12204 or QQ-B-654. For surfaces not likely to come into contact with moist food, the use of any of the commercial solders will be satisfactory.

3.4.15 Welding. When welded seams are used, the weld areas and deposited weld material shall be as resistant to corrosion as the parent material and shall conform to MIL-W-8604.

3.5 Design.

3.5.1 General design. The general configuration of type I and type II ovens shall be in accordance with figures 1 and 2. Type I ovens shall be designed to operate on 115/200 volt AC, 400 Hz electrical power and be 3 phase, 4 wire, "Y" connected grounded neutral. Type II ovens shall be designed to operate on 28 volt DC and be a 2 wire, ground power system. Both type I and II ovens shall have a minimum number of parts consistent with reliability. The designs, where practical, shall permit easy assembly, disassembly, location of trouble sources, and maintenance with tools and equipment normally available commercially. The entire assemblies shall provide maximum sanitation and ease of cleaning.

3.5.1.1 Insulation. Both type I and II ovens shall consist of a food warming chamber that is insulated on all six sides (an insulated door forming one side). The insulation on the top and the back of the chamber shall be covered with sheet-metal paneling that will also serve as a base for mounting electrical and other component parts requiring repairing or replacing. Insulation for the food chamber, insulation for the paneling, and the attaching parts shall in turn be mounted within a large outer case consisting of framework, sheet-metal paneling, an easily removable top cover, and an easily removable back panel. The insulating material shall be arranged so that it will remain in place throughout the life of the oven.

3.5.1.2 Motor, heating baffle, and fan. A motor shall be mounted on the back of the back wall under the removable back panel of both type I and II ovens and shall be attached to both the back wall and the back panel of the food warming chamber. The motor shaft shall extend through the back wall of the food warming chamber, and a fan shall be mounted on the inside of the back panel of the food warming chamber. A heating baffle containing an electrical resistance heating element shall fit over and around the fan in such a manner that air is drawn through a hole in the center of the baffle and then forced over the heating elements and around the sides of the sheet-metal plate holding the elements into the main part of the food warming chamber in which the metal rack is placed to hold meals to be heated. Jack type electrical connectors shall plug the heater baffle in and out of the back panel of the oven.

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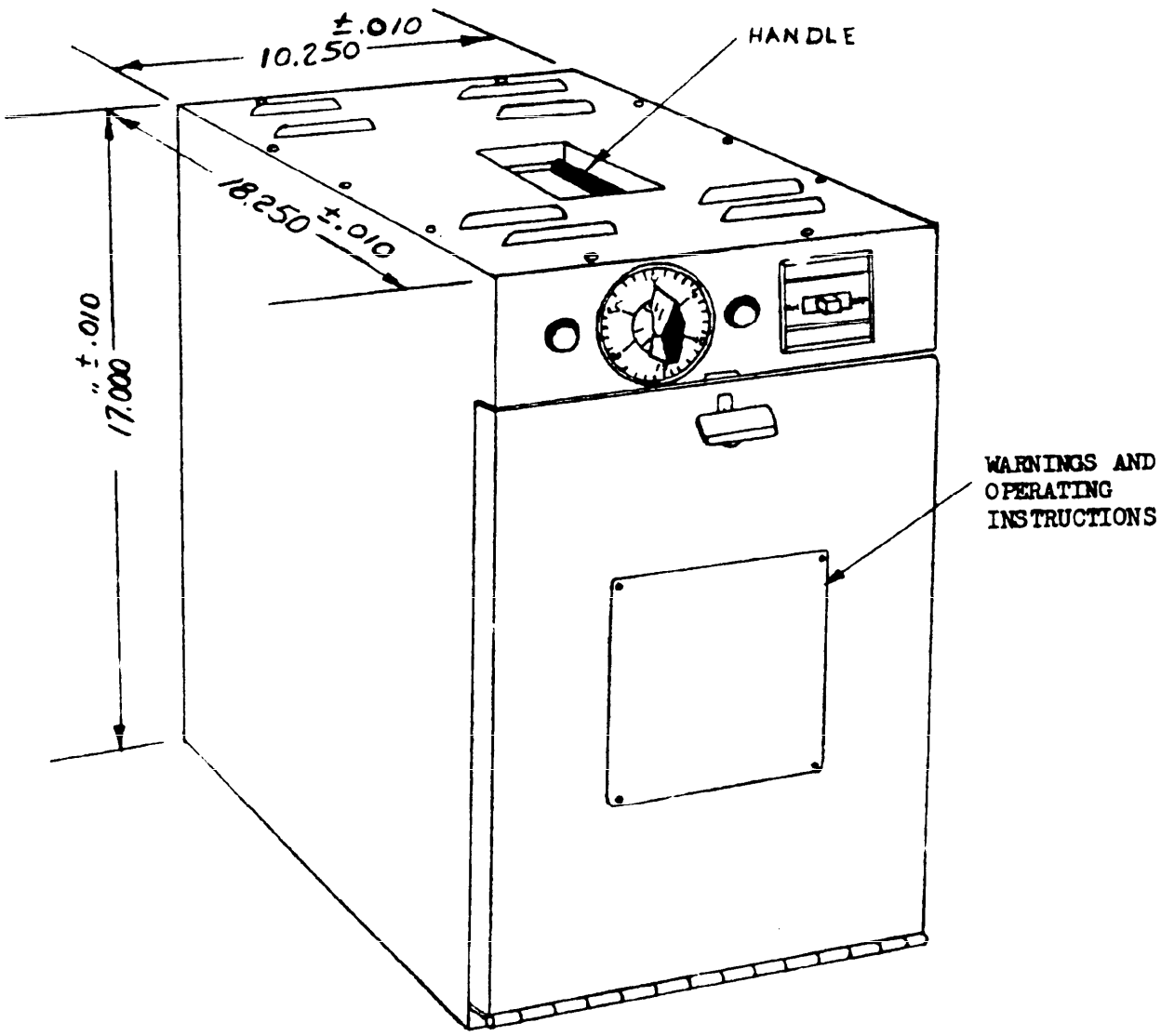


FIGURE 1

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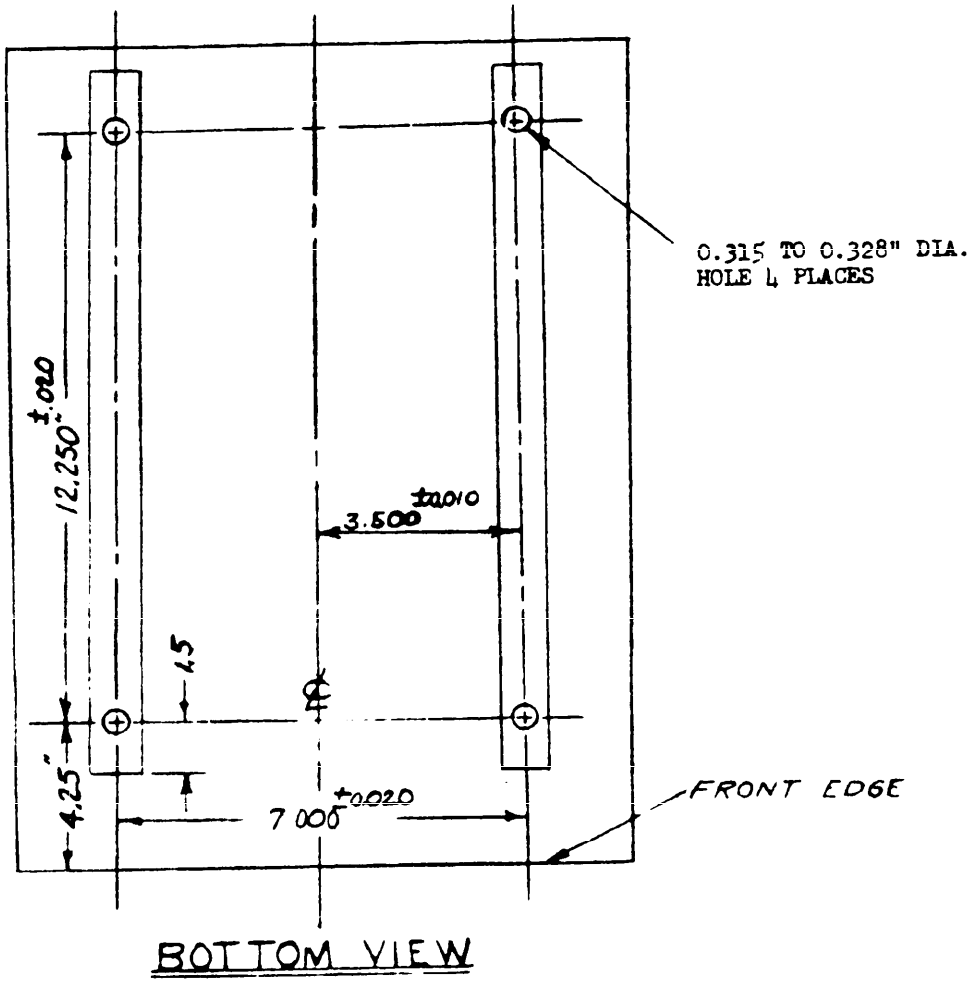


FIGURE 2

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3.5.1.3 Front panel. The front panel of the outer case shall extend above the door opening to the food warming chamber and shall serve as a mounting panel for a master switch, timer, red pilot light, and amber pilot light.

3.5.1.4 Electrical receptable. The electrical receptacles through which the power is supplied to the ovens shall be of a standard commercial design. Type I ovens shall have MS3106A20-4S type sockets and type II ovens shall have MS3102A20-23P type sockets.

3.5.1.5 Thermostat and timer. A fixed thermostat shall control the temperature of the heating chamber. Power shall cut off when the temperature of the chamber liner reaches 475° (plus or minus 10°) F. When the timer is set on any timing cycle and the master switch is placed in the "ON" position, power shall be supplied to the fan, red pilot light, heating elements, and amber pilot light. When the timer pointer has returned to the zero position and the master switch is in the "ON" position, power shall be supplied to the fan and the red pilot light but power shall be cut off to the heating elements and the amber pilot light. The master switch shall thus control power to the entire oven, and the timer shall be solely for the control of power to the heating elements and the amber pilot light. The red pilot light shall be on whenever the master switch is in the "ON" position and power is supplied to the oven. The amber pilot light is on only when the heating elements are operating. The fan shall operate whenever the master switch is in the "ON" position. All controls shall be immediately resettable for repeated operation on the oven. Fixed operating and fixed hi-limit thermostats shall be mounted on the rear insulation wall. A hi-limit capillary bulb shall be mounted on the top insulation wall. The oven shall be constructed in order to provide that if the operation thermostat should not function, the hi-limit will cut off the circuit to the heating element.

3.5.1.6 Carrying handle. An insulated carrying handle shall be attached to the top center area of the outer case of the oven in such a position that it will maintain a balance through the center-of-gravity of the oven. The place where the handle is located shall be recessed so that when the handle is not in use it will be even with the top of the oven. The handle shall be designed to support the entire weight of the oven.

3.6 Construction. The construction of the oven specified shall conform to figure 1.

3.6.1 Joints and seams. Joints shall be kept to a minimum and all surfaces shall be smooth and even. All seams shall be filled and smooth. Materials used to close seams shall bond to the metal so that it will not crack or chip off. If joints are welded, the welded areas shall be ground smooth. Seams and joints shall be closed to prevent fluids from entering the space between the inner and outer door panels and the space between the heating chamber and the cuter shell.

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3.6.2 Heating chamber. The sidewalls, the top, and the bottom of the heating chamber shall be constructed of one sheet of metal bent into shape and butt-welded so that the weld is along the top center of the chamber. The back of the chamber shall be constructed of one sheet of metal and shaped so that the corners have a minimum radius of 1 inch. The sidewalls, the top, and bottom shall then be butt-welded to the back. Internal angles within the heating chamber formed by the intersection of horizontal and vertical planes shall have a minimum continuous and smooth radius of $\frac{1}{4}$ inch. All exposed external angles shall be closed and smooth. The interior surfaces of the heating chamber, and any other surfaces within the chamber that could be exposed to spilled food and water, shall be free from projecting fasteners and exposed stud or screw threads. Fastenings within the heat chamber shall be designed to minimize projections and ledges. The interior surfaces of the heating chamber shall be smooth, continuous, and free from laps, protrusions, projections, ledges, cracks, voids, or depressions. Provisions shall be made for the expansion and contraction of all parts and surfaces in the heating chamber caused by heating and cooling, and no damage to the oven shall result from such thermal cycling. Two anchor receptacles shall be installed into back of the inner chamber and shall be locked into the baffle plate assembly.

3.6.3 Insulation, insulation cover, and mounting panels. The heating chamber shall be insulated with both rigid and nonrigid types of insulation material. A rigid type of insulation shall be used at the front of the heating chamber to facilitate sealing the drop-leaf door; all other areas shall be insulated with a nonrigid type of insulation. The insulation on the top and in back of the oven shall be covered by the electrical equipment mounting panel. Insulation shall also be placed between the heating chamber and the bottom and the sides of the outer case. It shall be held in place by an extension of the electrical equipment mounting panel. The insulation of the front of the oven shall be between the outer and inner door panels.

3.6.4 Outer case. The outer case shall completely surround the heating chamber, the insulation, the insulation panel, the electrical parts. The top shall be designed to permit easy access for repair and replacement of the parts mounted on the insulation cover panel. The removable back cover shall permit access to the fan motor and parts mounted on the back insulation wall panel. In the front of the oven the outer case shall be flush with the outer panel of the door. The controls shall be mounted in the area of the outer case above the door. Tongs shall be furnished with each oven and shall be stowed within the oven compartment for shipment to the using activities. Louvers for air circulation shall be located in the top and rear outer cover.

3.6.5 Mounting of manual controls and indicator lights. The master switch, the timing mechanism, the circuit breaker, the red pilot light, and the amber pilot light shall be mounted on the front of the oven.

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3.6.6 Heating elements and baffle. The heating elements shall be metal sheathed nickel chrome wire and shall be mounted on a baffle with the aid of adequate electrical insulators. The heating element and baffles shall be interchangeable among the same type ovens and shall be held in place by jacks that plug in place into receptacles mounted in the back wall of the heating chamber. Two anchor receptacles mounted in the back of the heating chamber shall lock into the baffle plate by means of two one-quarter turn lock screws, one on the bottom and one on the top. The two one-quarter turn lock screws shall be mounted in the baffle plate for quick lock into and release from the anchor receptacles.

3.6.7 Fan. The fan shall be of the centrifugal, backward curve blade type. The fan shall be easily removable from the motor shaft for cleaning. The fan shall be securely positioned on the motor shaft so that there will be no interference with the back wall or with the baffle plate. The fan shall be constructed of stainless steel, 18 gage minimum, type 304.

3.6.8 Drop-leaf door, door stay, hinge, catch, and strike. The drop-leaf door shall have a one-piece stainless steel inner liner. The liner shall be designed to eliminate seams on the inner face. It shall be folded around the side edge of the door and fastened to the edge. When the door is in the open position, this liner shall be usable as extra work space or as a platform for a rack of meals. A hinge or hinges with door linkage shall permit the door to open to a horizontal position. There shall be two door stays between the oven and the door so that the door will rest in a horizontal position when fully opened. Door stays shall project from and operate between the inner and outer wall of the oven. The door stay enclosure and reinforcement section shall be provided between the outer and inner walls of the oven. A pressed rim or some other means shall be provided to prevent the meal rack from sliding off the inner side of the opened door. The outer aluminum alloy cover of the door shall be thermally insulated from the inner liner (see 3.4.11). Provisions shall be made for the expansion and contraction of all parts and surfaces in the door assembly caused by heating and cooling, and no damage to the oven shall result from such thermal cycling. The door shall have a single-handle heavy duty, slam-shut type of door catch combined with a heavy duty strike, which shall be attached to the oven proper. The strike shall be adjustable to permit tightening of the door seal and to eliminate vibration of the door. The combination catch-strike shall provide a positive catch to eliminate accidental opening of the door. Provisions shall be made for effecting a tight seal between the door and the main body of the oven. When the oven is placed on a flat surface, the door handle shall not interfere with the opening of the door to the horizontal position. All projecting screws in the door-latch area shall protrude not more than 1/16 inch through the nuts. The door linkage shall incorporate adequate stops to prevent jack-knifing or hanging-up when the door is closed and shall provide for a positive stop when the door is open.

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3.6.9 Food rack. Each type oven shall have a food rack that is designed so that the following number and type of meals can be easily inserted, held, and removed: six precooked frozen Air Force platter or casserole type meals, six precooked frozen commercial "TV" platter type meals, six partially precooked and refrigerated foil pack meals, or 18 opened inflight feeding cans. The Air Force precooked frozen platter measures 8-½ inches in length, 6-½ inches in width, and 7/8 inch in depth (with food in the tray and the foil cover over the top, the total depth may vary up to 7/8 inch). For information concerning the Air Force precooked frozen meals, reference shall be made to MIL-M-13966 except for the type of container specified in the paragraph entitled "Trays and Casseroles". Each foil pack meal consist of one large foil pack container measuring 4-3/4 inches by 3-3/16 inches by 1-½ inches for four small foil pack containers each measuring 3-3/10 inches by 2-11/32 inches by 1-½ inches. Each inflight feeding can measures 3 inches in diameter by 2 inches in height. The rack shall be readily removable from the heating compartment.

3.6.10 Device for removing meal rack and meals. A device (tongs) shall be provided for removing the rack (with hot meals) from the oven.

3.6.11 Drip pan and auxiliary food tray. The drip pan and auxiliary food tray shall be designed to cover the bottom area of the heating chamber and shall be adaptable for use as an auxiliary food tray for bite size steaks. All edges of the drip pan shall be turned about ¼ inch to provide a container for drippings from spilled food. The pan shall be easily removed from the heating chamber for cleaning.

3.6.12 Mounting runners. Mounting runners shall be fitted to the underside of the ovens in accordance with figure 2.

3.7 Electrical requirements.

3.7.1 Electrical design. Electrical design of all wiring, devices, controls, and equipment for the ovens shall be compatible with the applicable design power source and shall be in accordance with MIL-STD-704. Type I ovens shall be designed to use a 115/200 volt, AC, 380-420 cycle, 3 phase, 4 wire "Y" connected, grounded neutral, electric power system. Type II ovens shall be designed to use 28 volt, DC, 2 wire, grounded, electric system.

3.7.2 Thermostatic control. Both type I and II ovens shall have a fixed temperature control that will maintain an oven liner temperature of plus 400° (plus or minus 10°) F. However, the temperature of the interior of the oven shall be set based on cooking six precooked Air Force frozen meals in a time interval of 22 minutes. The temperature of the food packages shall be approximately 5° F. when they are put into the oven and the temperature of the interior of the food shall be approximately plus 175° F. at the end of the prescribed length of time (see 3.8.3).

3.7.3 Safety high limit thermo-switch. The construction of the type I and II ovens, together with fixed thermo-switch control and a timer cut off switch, shall be such that the heat within a oven shall not be a hazard (see 3.8.2) in case an oven is turned on and utilized without a cooking load (empty) (see 4.6.3).

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3.7.4 Timing mechanism. The timing mechanism shall be an integral unit permitting manual selection of timing cycles between 5 and 30 minutes. When the timing mechanism is set to the desired time interval, power shall be delivered to the heating elements and amber light through the thermostat. When the set time has expired, power to the heating element and amber light shall automatically shut off indicating the end of the cycle. The timer shall be constructed so that movement of the dial in either direction will not damage the timer.

3.7.5 Pilot lights. One red pilot light to indicate that power is available to the oven and that the fan is operating (master switch "ON") shall be mounted on the front panel of the oven near the master switch. One amber pilot light to indicate that power is being supplied to the heating elements shall be mounted on the front panel near the timer.

3.7.6 Signal device. The amber pilot light for the timer operations shall signal when the meals are cooking. When the meal is cooked, the amber light shall shut off and the timer shall revert to the "OFF" position.

3.7.7 Master switch. A two-position switch shall be mounted on the front face of the oven. The down position shall be marked "OFF" and the up position shall be marked "ON". This switch shall control the power coming into the oven. When the master switch is in the "ON" position and the timer is at "O", power shall be supplied to the fan and the red light, but power shall not be supplied to the heating elements and the amber pilot light. When the master switch is in the "ON" position and the timer is set to any timing cycle, power shall be supplied on the fan, the red pilot light, the heating elements, and the amber pilot light. The circuit breaker may operate in either a horizontal or vertical position.

3.7.8 Heating elements (see 3.6.6). Type I and II ovens shall be equipped with a General Electric Company Calrod type unit, or equal, positioned heating element that is electrically insulated from the baffle plate. The insulation shall withstand rough treatment without damage to the insulating properties. The design service life of the heating element shall be not less than 2,000 hours of intermitent operations on a 30-minute cycle.

3.7.9 Motor. The motor shall be mounted so that it is easily removed for maintenance and replacement. The electrical characteristics of the motor shall conform to MIL-STD-704. The motor mount shall not permit excessive vibration. The motor to be used for circulating hot air in the type I oven shall be continuous duty type, 115/200 volt AC, 3 phase, 380-420 cycle, designed to operate at 10,000 revolutions per minute at 400 cycles. The AC power system shall be 3 phase, 4 wire "Y" connected, 115/200 volt, grounded-neutral (380-420 cycle) system. The motor to be used for circulating hot air in the type II ovens shall be a continuous duty type, designed to operate at 10,000 revolutions per minute at 28 volts DC.

3.7.10 Receptacles. For easy access to the power source, a male electrical receptacle shall be located at the top rear of the oven. The female half of the connector shall be delivered with the oven.

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3.7.11 Wiring and electromagnetic interference protection. All wiring and electrical connections for both type I and II ovens shall be in accordance with MIL-W-5088. The ovens shall conform to the electrical interference requirements of MIL-STD-461 for Class I A equipment. The electrical circuit of the ovens shall incorporate an adequate circuit breaker that will operate if the circuit becomes overloaded. This circuit breaker shall be mounted on the front panel of the oven where it is readily accessible. The circuit breaker and the master switch shall be combined into one DPDT circuit breaker.

3.7.12 Electrical insulation. The insulation shall be sufficient to enable the ovens to meet the dielectric strength requirements of 3.8.4. All electrical insulating materials shall incorporate moisture-resisting, temperature-resisting, strength, and dielectric properties necessary for the particular application of each material.

3.8 PERFORMANCE

3.8.1 Power consumption. The ovens shall be designed for minimum power consumption. The power consumption of the ovens shall not exceed 1,750 watts.

3.8.2 Safety high limit thermostat. The safety high limit fixed type thermostat shall cut off power to the ovens before the temperature of the outer case reaches 165° F.

3.8.3 Heating cycle (see 3.7.4). With the master power switch in the "ON" position, the red pilot light shall be lit, and the fan shall operate. When the timer is set to any given time (5 to 30 minutes) the amber light and the heating element shall operate. After the timer has returned to the zero position, the power to the amber light and the heating element shall be cut off, indicating the completion of the heating cycle. The red pilot light and the fan shall continue to operate until the master power switch has been turned off.

3.8.4 Dielectric strength. The ovens shall withstand without breakdown for a period of 1 minute the application of a 400 volt, 60 cycle, alternating potential between live metal parts and interconnected dead metal parts.

3.8.5 Heating capacity (see 3.7.3). With the ovens at an ambient temperature of 77° F., six precooked frozen meals (Beef Pot Roast, Menu No. 5 of MIL-M-13966 or a commercial "TV" dinner equal to Menu No. 5) that have been stored at minus 10° F. shall be heated to an internal (food) temperature of plus 175° F. within 22 minutes.

3.8.6 Outside surface temperature. The temperature of the outside surfaces shall not exceed plus 165° F. during the cooking cycle.

3.8.7 Accuracy of timer. The accuracy of the timer shall be within plus or minus 30 seconds based on a 30 minute cycle.

3.9 Weight. The weight of the oven shall be held to a minimum consistent with configuration, strength, and other requirements specified herein. The weight shall not exceed 30 pounds.

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3.10 Dimensions. The dimensions shall be 17 inches in height, 10¾ inches in width, and 18¾ inches in depth nominally.

3.11 Finish. All external sheet metal surfaces shall have a commercial polish finish. Heating chamber interior surfaces shall have a 180 grit finish.

3.12 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Instructions. Operating instructions shall be placed on the outer door panel. Warning instructions shall be attached to the outside door panel. The warning instructions shall precede the operating instructions. The warning instructions shall be as follows:

WARNING (¼ inch minimum letters)

DO NOT HEAT CANNED FOOD IN THIS OVEN WITHOUT

FIRST PUNCTURING OR REMOVING END OF EACH CONTAINER.

The operating instructions shall be as follows:

OPERATING INSTRUCTIONS (3/16 inch minimum letters)

1. Turn on oven and allow 5 minutes preheat time.
2. Open all canned food prior to heating.
3. Place meal containers in food rack or add shaped cans on the floor of the heating chamber. Close door tightly.
4. Set desired cooking time.
5. Move master switch to "ON" position.
6. When heating time expires, power to heating element will automatically shut off. (Power will continue to fan and red pilot light until master switch is returned to "OFF" position).
7. Make sure covers of precooked frozen and foil pack meals are tightly in place to prevent drying out. Cover open inflight feeding cans with aluminum foil, crimping foil down along sides of each container.

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3.15 Workmanship. The ovens shall be neatly finished and all burrs or rough edges shall be removed. Welds shall be sound, smooth and free from pitts, fractures, burn through or fissures. Bolt holes and rivet holes shall be accurately punched or drilled and shall have burrs removed. Lockwashers shall be provided in accordance with standard commercial practice, and all bolts, nuts and screws shall be drawn tight. Rivets shall be tightly driven and fully peened. Electrical wiring shall be provided with adequate slack to relieve strain, and the wiring shall not be abraded, poorly joined, or loose at terminals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the ovens shall be classified as follows:

a. Preproduction testing.

b. Acceptance tests.

4.3 Test conditions.

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77 degrees (plus or minus 10 degrees) Fahrenheit.

4.3.2 Electrical power. All tests of the type I oven shall be conducted using 115/200 volts, 380-420 cycle, 3 phase, AC current, four wire "Y". All tests of the type II oven shall be conducted using 28 volt DC current except for the test specified in 4.6.5.

4.4 Preproduction testing.

4.4.1 Preproduction test sample tested by the contractor. The contractor shall subject one oven to the preproduction tests specified in 4.4.2.

4.4.2 Preproduction tests. Preproduction tests shall consist of all test described under 4.6 Test Methods.

4.5 Acceptance tests. Acceptance test shall consist of:

a. Individual tests

b. Sampling tests

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4.5.1 Individual tests. Each oven shall be subjected to the following tests as described under 4.6 Test Methods.

a. Examination of product (4.6.1).

b. Power consumption (4.6.2).

c. Safety high limit thermostat (4.6.3).

d. Heating cycle (4.6.4).

e. Dielectric strength (4.6.5).

4.5.2 Sampling tests. One oven shall be selected at random from each lot of 100 or fraction thereof and subjected to the following tests as described under 4.6 Test Methods.

a. Heating capacity (4.6.6).

b. Outside surface temperature (4.6.7).

c. Accuracy of timer (4.6.8).

d. Electromagnetic interference control (4.6.10).

4.5.2.1 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot will be withheld until the extent and cause of failures are determined. The contractor shall explain fully to the Government representatives the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.5.2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test method.

4.6.1 Examination of product. Unless otherwise specified by the procuring activity, each oven shall be examined to determine conformance to this specification with respect to material, workmanship, dimensions, weight, markings, and excessive vibration of the motor, fan, and door.

4.6.2 Power consumption. The power consumption of each element and the entire oven shall be determined with a wattmeter or any other suitable means to determine compliance with 3.8.1.

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4.6.3 Safety high limit thermostat. The safety high limit thermostat shall be tested to determine if it will cut off power to the oven before the temperature of the outer case reaches 165° F. The oven shall be turned on and cycled without any load (empty).

4.6.4 Heating cycle. The master power switch shall be placed in the "ON" position, and a check shall be made to determine if the red pilot light is lit and if the fan is operating. The timer shall be set, and a check shall be made to insure that electrical power is supplied to the heating element and the amber light. While the timer is in operation, a visual check shall be made to determine if the amber and red lights are lit. When the timer reaches zero, a visual check shall be made to determine if the amber light is off and if the power to the heating element is off. The master power switch shall be turned off and shall be checked to determine if the red pilot light is off and power is not being supplied to the fan.

4.6.5 Dielectric strength. After final assembly, the output terminals of a suitable transformer, the output of which can be regulated, shall be connected to the oven case and a jumper which interconnects all terminals of the plug. The potential shall be increased gradually until a test voltage of 400 volts is reached without breakdown occurring.

4.6.6 Heating capacity. The meals shall be placed in the oven while the oven is at standard room temperature. The power to the oven shall be turned on and the internal (food) temperature shall be recorded at the end of the heating cycle. At the conclusion of this test, the controls shall immediately be reset, and the lights and the timer shall be checked for proper operation (see 3.5.1.5).

4.6.7 Outside surface temperature. The temperature of the outside surface of the oven shall be determined on each face of the oven. The temperature shall not be greater than plus 165° F. This test may be made in conjunction with the test specified in 4.6.4.

4.6.8 Accuracy of timer. Test under normal operating conditions and in accordance with para 3.8.7 of this specification.

4.6.9 Reliability. The oven shall be properly connected and subjected to a cycle consisting of 30 minutes of operation at the maximum required temperature of the oven in accordance with 4.6.4. The heating cycle shall be repeated a sufficient number of times to permit acceptance or rejection in accordance with table I. Operating hours accumulated during the other tests required may be considered as part of the reliability test time. In the event of a rejection all shipments shall be discontinued until an evaluation is made, corrective action taken, and a new item has been subjected to and has successfully passed the reliability test. All out of tolerance conditions shall be considered as failures. Each performance parameter shall be checked at least once during each 24 hour test period and at the completion of the reliability test.

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TABLE I - ACCEPT-REJECT CRITERIA

FAILURES	TOTAL TEST TIME IN OPERATING HOURS
0	230
1	390
2	530
3	670
Over 3 1/	

1/ Equipment shall be rejected subject to redesign as required to demonstrate the required reliability.

4.6.10 Electromagnetic interference control. Both type I and II ovens shall be tested for compliance with 3.6.11 in accordance with table II of MIL-STD-461.

4.6.11 Vibration test. The oven, properly connected and operating normally, shall be subjected to a vibration test. The oven shall be mounted on the vibrating apparatus in the horizontal position. The oven shall be vibrated with circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45° to the horizontal and the frequency of vibration shall be varied uniformly from 5 to 50 cycles per second and return once each hour for a 3-hour period. At the completion of this test, the oven shall be operated and the results compared to the data obtained in 4.6.4. A visual inspection shall be in accordance with 4.6.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each oven shall be preserved in accordance with Method IA of MIL-P-116 and shall be packaged in a box conforming to type I or II, class 1 or 2 of PPP-B-636.

5.1.2 Level C. Level C packaging shall be in accordance with manufacturers' commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Ovens preserved and packaged as specified in 5.1.1 shall be packed in boxes conforming to type II, class 2 of PPP-B-636.

5.2.2 Level B. Ovens preserved and packaged as specified in 5.1.1 shall be packed in boxes conforming to type I, class 2 of PPP-B-636.

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5.2.3 Level C. Ovens shall be packed in a manner that will afford adequate protection against corrosion, deterioration, and physical damage and shall conform to applicable rules of carriers and regulations.

6. NOTES

6.1 Intended use. The ovens covered by this specification are intended to be used to heat Air Force and commercial "TV" and precooked frozen meals, partially precooked and refrigerated foil pack meals, and opened inflight feeding cans in aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type I or type II oven required.

c. List of interchangeable components and assemblies.

d. Selection of applicable levels of packaging and packing.

6.3 Supersession data. This specification includes the requirements of MIL-O-38246 and MIL-O-38205, both dated 6 April 1964. MIL-O-38246 has been cancelled.

6.4 Marginal notations. The margins of this specification were not marked with asterisks to indicate changes from the previous issue due to the extensiveness of the changes.

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Air Force - 84

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